SECI Knowledge Conversion Model and Tacit Transmission Loss (Nonaka 1994)

Organizational knowledge creation requires continuous conversion between tacit and explicit dimensions across four modes (Socialization, Externalization, Combination, Internalization); restructuring disrupts the fragile tacit socialization phase.

GRADE: MODERATE | STATUS: DRAFT | DOMAIN: ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Organizational knowledge is created through dynamic interactions between tacit knowledge (subjective, experiential, context-specific insights that are difficult to formalize) and explicit knowledge (codified, systematic language and data). This conversion operates through a continuous spiral spanning four modes: Socialization (tacit to tacit via shared experience), Externalization (tacit to explicit through conceptual modeling), Combination (explicit to explicit through systemic sorting and synthesis), and Internalization (explicit to tacit through learning-by-doing). When organizations execute sudden structural reorganizations or sever apprentice-mentor relationships, the socialization phase is broken, causing irreversible loss of tacit knowledge that cannot be recovered through documentation alone.

EVIDENCE AND CAUSAL ASSESSMENT

Nonaka (1994) formalized the epistemological and ontological dimensions of organizational knowledge creation using extensive field evidence from Japanese manufacturing innovation cycles (e.g., the Matsushita home bakery project). Nonaka demonstrated that codified documentation (Combination) is ineffective unless grounded in shared tacit models (Socialization) and conceptual metaphors (Externalization). Field studies in aerospace engineering (e.g., NASA post-Challenger and post-Columbia investigations) corroborate that when informal, experiential socialization channels are dismantled by organizational churn, critical failure indicators cease to be recognized by decision-makers.

EFFECT SIZE AND PERSISTENCE

Tacit knowledge transfer requires prolonged interactive exposure (typically months to years of collaborative co-location). Reorganizations that shuffle cross-functional teams reset the socialization cycle, resulting in immediate productivity dips of 20% to 40% in complex technical domains before new shared informal networks form.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as partially confounded (20–50%) (tacit knowledge preservation correlates with organizational culture, tenure stability, psychological safety, and apprenticeship structures).

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

This model does not imply that explicit documentation is useless, nor does it justify mystical or un-auditable corporate cultures that resist formalization; tacit knowledge must be externalized into shared frameworks to become scalable across an enterprise.

WHAT ACTION IS REASONABLY JUSTIFIED

Enterprises undergoing technical transitions or structural reorganization should pair formal procedural documentation with protected apprentice-mentor shadowing programs, maintain persistent cross-functional team cores, and avoid treating codified wikis as complete substitutes for relational knowledge networks.

VERIFIED SOURCES & DOI IDENTIFIERS

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- Walsh, J. P., & Ungson, G. R. (1991). Organizational Memory. The Academy of Management Review, 16(1), 57–91. https://doi.org/10.2307/258607